

not solely on the disclosure of the closest prior art document, but also on the state of the art in the relevant technical field.

In T. 1126/09 the board pointed out that, in accordance with the "could-would" approach, the assessment of inventive step must involve establishing, in each individual case, to what extent the skilled person had good reason, in the light of the closest state of the art or the objective problem derivable from it, to adduce further prior art and apply its teaching to the process/apparatus of the closest prior art – or, in other words, whether any factor is discernible which points towards a combination of the teachings of the citations addressed.

Technical feasibility and the absence of obstacles were only necessary requirements for reproducibility but were not sufficient to render obvious **what was actually achievable** for the skilled person (T. 61/90). The fact that the inherent properties of a technical means were known to the skilled person, so that he had the intellectual possibility to apply this means in a conventional device, merely established the **possibility** of using such technical means in such a manner, i.e. that the skilled person **could** have used it. However, if it was to be established that such intellectual possibility was also a technical measure which it was obvious for the skilled person to use, it was necessary to show that there was a recognisable pointer in the state of the art to combine the known means and conventional device for achieving the intended technical aim, i.e. that the skilled person **would** have made such a combination. The existence of such a technical reason was dependent on the known properties not only of the means but also of those of the device (T. 203/93, T. 280/95). That it was theoretically possible for the skilled person to arrive at the invention simply means that he could have used the requisite technical means. If, however, it is to be established that he would actually have used them, it must be possible to ascertain a pointer in the prior art which would have prompted him to do so (T. 1317/08).

In T. 905/17 the board did not find persuasive the opposition division's dismissal of a line of argument on the basis that there was no "hint in document D4 to suppress the liner (3) and then to search for appropriate material in the prior [sic] for the pressure armours (5,6) then subjected to corrosion". The skilled person starting from one element of the state of the art and faced with the need to solve a given problem did not necessarily need a "hint" associated with that element. Otherwise, it would never be possible to establish a lack of inventive step based on the object of a prior public use, which usually does not come with any hints. In the absence of a hint, the skilled person might still take the steps leading to the claimed subject-matter on the basis of their common general knowledge or documents belonging to the state of the art that explicitly teach a solution to the problem to be solved.

In T. 894/19 the board noted that in cases such as the case in hand, whereby the configuration of features represented a mere obvious and consequently non-inventive selection among a number of known and equally likely possibilities, the "could-would approach" normally did not apply (see also T. 1968/08, T. 12/07). It held that to consider all of the solutions that were equally obvious, it was sufficient that the skilled person could recognise the solutions concerned without inventive efforts: a separate pointer was then not required for this purpose.

6. Ex post facto analysis

Many decisions of the boards of appeal warn against an ex post facto approach when assessing inventive step (see also the Guidelines G-VII, 8 – March 2022 version). This applies especially to inventions which at first sight seem obvious, to combination inventions and where the proposed solution is supposedly "simple". Correct application of the problem and solution approach avoids this inadmissible ex post facto analysis which draws on knowledge of the invention (T 24/81, OJ 1983, 133; T 564/89; T 645/92; T 795/93).

When assessing inventive step, an interpretation of the prior art documents as influenced by the problem solved by the invention, where the problem was neither mentioned or even suggested in those documents, must be avoided, such an approach being merely the result of an a posteriori analysis (T 5/81, OJ 1982, 249; T 63/97; T 170/97; T 414/98).

In T 970/00 the board stated that any ex post facto analysis, and in particular any conclusion going beyond what the skilled person would have objectively inferred from the prior art, without the benefit of hindsight knowledge of the invention, is of necessity at variance with a proper application of the problem and solution approach. Any attempt to interpret the disclosure of the closest prior art so as to distort or misrepresent, based on hindsight knowledge of the invention, the proper technical teaching of the disclosure in such a way that it artificially meets specific features recited in the claim under consideration must fail, especially as this would risk unfairly and tendentiously concealing the technical contribution of the invention and prejudice the subsequent objective determination of the technical problem solved by the claimed invention (see also T 266/07, T 1486/10).

In T 2201/10, the board found that the examining division had taken an ex-post-facto approach to analysing the facts. Even assuming that the proposed solution could be derived from common general knowledge, it went against the very essence of the teaching in the closest prior art and, therefore, would not realistically have been chosen. The problem and solution approach required that documents not relating to the technical field of the invention be disregarded, and it also precluded any analysis according to which the skilled person would have modified the closest prior art in a manner contrary to its very purpose. In other words, a finding that a claimed invention deviated from what was, in light of the aim pursued, the essence of a prior-art disclosure was enough by itself to conclude that it was inventive over that prior art (see also T 2057/12).

In T 855/15 the board considered that for the assessment of obviousness a consideration as to what the skilled person would do, in particular whether the skilled person "would select" a particular document in order "to arrive at the invention as claimed" would amount to hindsight reasoning, because the skilled person would have to be assumed to know the invention before an argument could be made as to what he would do in order "to arrive at" it. The board in T 2057/12 agreed with the board in T 855/15 insofar as the closest prior art belonged to the same or a neighbouring technical field of the skilled person or to his common general knowledge. Argumentation or evidence should be provided as to why the skilled person in a specific technical field would have indeed envisaged selecting a document in a remote field of technology as closest prior art or whether he would indeed

have considered adapting a prior-art disclosure originating from his technical field to implement it in a remote technical field.

In T 1087/15 the board pointed out that knowledge of the claimed invention was absolutely necessary in order to formulate the objective technical problem, irrespective of how 'close' the prior art document was to the claimed invention; the features differentiating the claimed subject-matter from the starting document must be identified, their technical effect determined and thus the problem to be solved formulated. In as far as such knowledge of the claimed invention was labelled as 'hindsight', this particular type of hindsight was indeed a necessity in order for the technical problem to be formulated on an objective basis. As long as there was no reason (primarily a technical reason) why the content of the document selected as the closest prior art would itself not be suitable or compatible for use with respect to the subject-matter claimed, any analysis of 'how close' the prior art starting document was judged to be to the claimed invention was not something which should exclude it from being the closest prior art.

7. Expectation of success, especially in the field of genetic engineering and biotechnology

7.1. Reasonable expectation of success

In accordance with the case law of the boards of appeal, a course of action can be considered obvious within the meaning of Art. 56 EPC if the skilled person would have carried it out in expectation of some improvement or advantage (T 2/83, OJ 1984, 265). In other words, obviousness is not only at hand when the results are clearly predictable but also when there is a reasonable expectation of success (T 149/93). It is not necessary to establish that the success of an envisaged solution of a technical problem was predictable with certainty. In order to render a solution obvious it is sufficient to establish that the skilled person would have followed the teaching of the prior art with a reasonable expectation of success (T 249/88, T 1053/93, T 318/02, T 1877/08, T 2168/11, T 867/13).

In some decisions, especially in the field of biotechnology, the boards have asked whether in the cases in point it was obvious for the skilled person to try a suggested approach, route or method with a **reasonable expectation of success** (T 60/89, OJ 1992, 268). For more about biotechnological inventions and the definition of the skilled person, see also point I.D.8.1.3 below.

In T 2168/11 the board referred to the case law, according to which the expectation of success depended on the complexity of the technical problem to be solved. While for very ambitious problems requiring the consideration of all the features relied on by the respondent (patent proprietor) but not contained in claim 1, important difficulties might be expected a priori, less ambitious problems might normally be associated with higher expectation of success (see T 192/06, T 782/07, T 688/14, T 967/16).

In line with T 918/01, the board in T 1577/11 concluded that, given the superior efficacy of anastrozole, as compared with tamoxifen, in treating advanced breast cancer, there was

a reasonable expectation it would also improve the treatment of early breast cancer, as compared with that achieved with tamoxifen.

In T.1680/17 the board considered that the skilled person would not have turned to formulations used in basic research when aiming at providing a formulation for therapeutic treatment. The requirements of a formulation to be used in basic research were fundamentally different from the requirements of a formulation to be administered to a patient.

In T.296/93 the board held that, in relation to inventive step, the fact that other persons or teams were working contemporaneously on the same project might suggest that it was "obvious to try" or that it was an interesting area to explore, but it did not necessarily imply that there was a "reasonable expectation of success". A reasonable expectation of success should not be confused with the understandable "hope to succeed"; it implied the ability of the skilled person to predict rationally, on the basis of the knowledge existing before a research project was started, the successful conclusion of the said project within acceptable time limits. The more unexplored a technical field of research was, the more difficult it was to make predictions about its successful conclusion and, consequently, the lower the expectation of success (T.694/92, OJ 1997, 408). According to T.207/94 (OJ 1999, 273), the "hope to succeed" was merely the expression of a wish, whereas a "reasonable expectation of success" presupposed scientific appraisal of available facts.

In T.187/93 it was stated that even if it was obvious for the skilled person to try an experiment, it was not necessarily true that this person would have any reasonable expectation of success when embarking on it.

In T.223/92 the board said that in 1981, given the state of the art at that time, the skilled person would have opted for DNA-recombination technology only if relying, e.g., on his own good luck and inventiveness to overcome the known (and as yet unknown) problems involved, which would have caused the average skilled person to expect to fail.

In T.923/92 (OJ 1996, 564) the board had to decide whether the skilled person would have attempted, with reasonable expectation of success, to produce cDNA coding for human t-PA, or whether in this instance he would have known from his technical knowledge, before even embarking on the research, that he would be able to complete his project within an acceptable time. The board bore in mind that, as stated in T.816/90, even when it was possible to theoretically conceive a straightforward approach to solve a specific technical problem, the skilled person might be confronted with unexpected difficulties when trying to put the conceived strategy into practice. The board stated that, although hoping to succeed, the skilled person embarking on this project would have known that its successful conclusion depended not only on technical skill in putting into practice the sequence of precise steps of the theoretical experimental protocol, but to a large extent also on the ability to take the right decisions along the way whenever a difficult experimental situation so required. Under these circumstances, it could not be said that the skilled person had a reasonable expectation of success.

In T.386/94 (OJ 1996, 658), again citing T.816/90, the board ruled that, in gene technology, inventive step could not be acknowledged if, at the priority date, a skilled person could expect to perform the cloning and expression of a gene in a fairly straightforward manner, and the cloning, although requiring much work, did not pose such problems as to prove that the expectation of success was ill-founded.

In T.207/94 (OJ 1999, 273) the board held that where the expression of a cloned DNA in a chosen foreign host constituted the subject-matter of the claimed invention, the question whether a reasonable expectation of success existed or not could be evaluated only by taking into account real difficulties relating to that step. Thus, in order to be considered, any allegation that features jeopardised a reasonable expectation of success had to be based on technical facts.

In T.737/96 the board was of the opinion that it was not appropriate to attempt to evaluate the expectation of success of a random technique such as mutagenesis where results depended on chance events. This was because the skilled person knew that, unless a specific selection method could be developed, which was not the case in the patent in suit, perseverance and chance played a key role in achieving success, as no form of control could be exerted over the mutation events. Under these circumstances, as in a lottery game, the expectation of success always ranged irrationally from nil to high, so it could not be evaluated in a rational manner based on technical facts (see also T.694/92, OJ 1997, 408).

In T.967/16 the board found that there was no requirement in the method of claim 1 as regards the predictive value of the correlation between the HLA-B*1502 allele and the adverse drug reaction SJS/TEN developed in response to, or associated with, the OXC or LIC treatment. In other words, the expectations of a skilled person for the value of this correlation could be as high as 100% but also, and more importantly, as low as 47% or even lower than those reported in document (4) for other aromatic anticonvulsants less, or not at all, related to CBZ. In this sense, the board found that the formulated technical problem was not very ambitious and thus, the skilled person had a high expectation of success.

In T.96/20 the board considered that the announcement of a detailed safety and efficacy clinical trial protocol for a particular therapeutic and disease provided the skilled person with a reasonable expectation of the success of this particular therapeutic, unless there was evidence to the contrary in the state of the art. The board failed to see how the mere fact that no myasthenia gravis therapy had been approved for a long time would have diminished the expectation of success for the specific clinical trial disclosed in the prior art.

7.2. Try and see situation

When neither the implementation nor the testing of an approach suggested by the prior art involves any particular technical difficulties, the consideration that the skilled person would have at least adopted a "try and see" attitude is a reason for denying inventive step (see e.g. T.333/97, T.377/95 of 24 April 2001, T.1045/98, T.1396/06, T.2168/11). In such situations the concept of "reasonable expectation of success" does not apply (T.91/98,